

ACM SUMMER SCHOOL ON SYMMETRIC KEY CRYPTOGRAPHY (08-17 June, 2026)

Date	Time	Topic	Speaker(s)
8-Jun-2026 (Mon)	10:00-12:00	Introduction to cryptography. Historical Ciphers. Information-theoretic secrecy. Computational indistinguishability.	M V Panduranga Rao
	12:00-1:00	LUNCH	
	1:00-2:00	Introduction to cryptography. Historical Ciphers. Information-theoretic secrecy. Computational indistinguishability.	M V Panduranga Rao
	2:00-3:00	Attacks on Historical Ciphers	Nitin Sharma, TA: Ankit Agarwal and Sudhir Panda
	3:00-3:30	TEA BREAK	
	3:30-5:00	Attacks on Historical Ciphers	Nitin Sharma, TA: Ankit Agarwal and Sudhir Panda
9-Jun-2026 (Tue)	10:00-12:00	Pseudorandom Generators (PRGs), Pseudorandom Functions (PRFs), Pseudorandom Permutations (PRPs), Birthday Paradox, PRF/PRP Switching Lemma	Maria Francis
	12:00-1:00	LUNCH	
	1:00-2:00	Pseudorandom Generators (PRGs), Pseudorandom Functions (PRFs), Pseudorandom Permutations (PRPs), Birthday Paradox, PRF/PRP Switching Lemma	Maria Francis
	2:00-3:00	Building Distinguishers for PRFs	Preethi Thakkalapally. TA: Jayant More and Saswata Mishra
	3:00-3:30	TEA BREAK	
	3:30-5:00	Building Distinguishers for PRFs	Preethi Thakkalapally. TA: Jayant More and Saswata Mishra
10-Jun-2026 (Wed)	10:00-12:00	Block Ciphers: Feistel Networks, SPNs.	M. V. Panduranga Rao
	12:00-1:00	LUNCH	
	1:00-2:00	DES and AES Construction	M. V. Panduranga Rao
	2:00-3:00	DES and AES Encryption/Decryption Programs	Preethi Thakkalapally. TA: Vidya Praveen and Sudhir Panda
	3:00-3:30	TEA BREAK	
	3:30-5:00	DES and AES Encryption/Decryption Programs	Preethi Thakkalapally. TA: Vidya Praveen and Sudhir Panda
	10:00-12:00	Linear Cryptanalysis and Its Improvements	Preethi Thakkalapally

11-Jun-2026 (Thu)	12:00-1:00	LUNCH	
	1:00-2:00	Linear Cryptanalysis and Its Improvements	Preethi Thakkalapally
	2:00-3:00	Hands-on Linear Cryptanalysis on DES	Preethi Thakkalapally. TA: Vidya Praveen, Ankit Agarwal
	3:00-3:30	TEA BREAK	
	3:30-5:00	Hands-on Linear Cryptanalysis on DES	Preethi Thakkalapally. TA: Vidya Praveen, Ankit Agarwal
12-Jun-2026 (Fri)	10:00-12:00	Differential Cryptanalysis and Its Enhancements	Sabyasachi Dey
	12:00-1:00	LUNCH	
	1:00-2:00	Differential Cryptanalysis and Its Enhancements	Sabyasachi Dey
	2:00-3:00	Hands-on Differential Cryptanalysis on DES	Nitin Sharma. TA: Sawata Mishra, Hrushikesh Gaddam
	3:00-3:30	TEA BREAK	
	3:30-5:00	Hands-on Differential Cryptanalysis on DES	Nitin Sharma. TA: Sawata Mishra, Hrushikesh Gaddam
13-Jun-2026 (Sat)	10:00-12:00	Introduction to Stream Ciphers, LFSR and ARX	Virendra Sule
	12:00-1:00	LUNCH	
	1:00-2:00	Introduction to Stream Ciphers, LFSR and ARX	Nitin Sharma
	2:00-3:00	Cryptanalysis of Stream Ciphers	Nitin Sharma. TA: Hrushikesh Gaddam, Jayant More
	3:00-3:30	TEA BREAK	
	3:30-5:00	Cryptanalysis of Stream Ciphers	Nitin Sharma. TA: Hrushikesh Gaddam, Jayant More
14-Jun-2026 (Sun)	10:00-12:00	Introduction to Side-Channel Attacks	Shirshendu Das
	12:00-1:00	LUNCH	
	1:00-2:00	Introduction to Side-Channel Attacks	Shirshendu Das
	2:00-3:00	Hands-on Side-Channel Attacks: Cache and Timing Attacks	Shirshendu Das
	3:00-3:30	TEA BREAK	
	3:30-5:00	Hands-on Side-Channel Attacks: Cache and Timing Attacks	Shirshendu Das TA:

ADVANCED TOPICS IN CRYPTOGRAPHY

15-Jun-2026 (Mon)	10:00-12:00	Cryptanalysis using Machine Learning/Deep Learning Techniques	Nitin Sharma
	12:00-1:00	LUNCH	
	1:00-2:00	Advanced Topics in Cryptography-I	Dhiman Saha
	2:00-3:00	Advanced Topics in Cryptography-II	Shibam Ghosh
	3:00-3:30	TEA BREAK	
	3:30-5:00	Advanced Topics in Cryptography-II	Shibam Ghosh
16-Jun-2026 (Tue)	10:00-12:00	Advanced Topics in Cryptography-III	C. Pandu Rangan
	12:00-1:00	LUNCH	
	1:00-2:30	Theorem proving in Lean	Ashish Mishra
	2:30-3:00	TEA BREAK	
	3:00-5:00	Cryptography for Enterprises-PQC Crypto agility, Cryptography for Enterprises-Privacy Preserving Machine Learning	TCS
17-Jun-2026 (Wed)	10:00-12:00	Cryptography for Enterprises-Future of Payments (CBDC)	TCS
	12:00-1:00	LUNCH	
	1:00-2:00	Cryptography for Enterprises-Future of Payments (CBDC)	TCS
	2:00-3:00	Cryptography for Enterprises-Verifiable credentials	TCS
	3:00-3:30	TEA BREAK	
	3:30-5:00	Cryptography for Enterprises-Verifiable credentials	TCS